POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT III: Diesel Power Plant and Gas Turbine Plant

1. Diesel Power Plant – Introduction

- A ***diesel power plant*** uses a ***diesel engine (compression-ignition, CI)*** as prime mover to drive an alternator. Fuel is burnt inside the cylinder (internal combustion).
- Rating: from a few kW to about ***50 MW***; typical stations 2–50 MW. Used as ***stand-by***, ***peak-load***, ***emergency*** and ***isolated / remote area*** supply (hospitals, telecom, islands, mines, industries).

Advantages

- Simple design and layout, ***quick start*** (in ~ 1–2 minutes) and stop, small space, low initial cost, ***no stand-by losses***, thermal efficiency 35–40% (higher than a small steam plant), little cooling water, easy to operate, good part-load efficiency.

Disadvantages

- ***High running cost*** (diesel is costly), high maintenance and lubricating oil cost, limited capacity, ***noise*** and vibration, smoky exhaust (NO_x, soot), cannot take overloads for long, fuel storage hazards.

2. Working of Diesel Engine (revision)

- Air alone is ***compressed*** (compression ratio ***14:1 – 22:1***) -> temperature rises to ~ 500–700 deg C; fuel is injected as a fine spray at the end of compression and ***self-ignites*** (no spark plug).

Point	4-stroke engine	2-stroke engine
Cycle completed in	4 strokes (2 revolutions)	2 strokes (1 revolution)
Strokes	Suction, compression, power, exhaust	Compression + (suction / scavenging) combined with power + exhaust
Valves	Inlet + exhaust valves	Ports (scavenge + exhaust)

Point	4-stroke engine	2-stroke engine
Power	Less for same size	Nearly double for same size
Efficiency	Better fuel economy	Some fuel lost in scavenging
Use	Medium / small sets	Large, slow marine / power engines

- Cycle: ideal **Diesel cycle** (heat added at constant pressure); real engines follow the **dual cycle**.

> Diesel cycle efficiency = $1 - (1 / r^{gamma-1}) \times [(rho^{gamma} - 1) / (gamma (rho - 1))]$ (r = compression ratio, rho = cut-off ratio)

- **Brake thermal efficiency** = brake power / (fuel flow x calorific value). **Specific fuel consumption** (sfc) = about **0.20–0.25 kg/kWh**.

3. Plant Layout with Auxiliaries

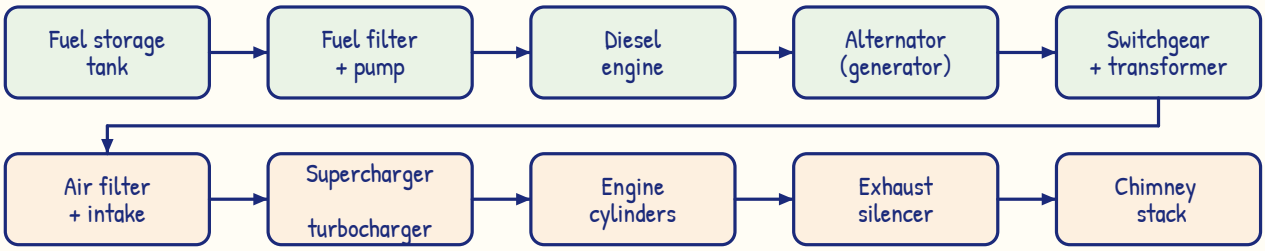


Fig: Block diagram of a diesel power plant

Main parts

- **Diesel engine** (prime mover) coupled directly to an **alternator** (synchronous), usually 3-phase, 415 V / 11 kV, 1500 rpm (4 poles) or 750 rpm.
- Foundations: heavy, **vibration-isolated** concrete block with a gap to the building floor.
- **Control room** with switchgear, **governor** (maintains constant speed), AVR (voltage regulator), protection panels.

Auxiliary systems

System	Function
Fuel supply system	Storage tank (bulk) -> transfer pump -> day tank -> filter -> **fuel injection pump** -> injector (atomiser, 150-300 bar)
Air intake system	Air filter (dry / oil bath), intake manifold, **supercharger / turbocharger** to raise power by 30-50%
Exhaust system	Exhaust pipe, **silencer / muffler**, waste heat boiler, chimney; reduces noise and carries gases outside
Cooling system	Jacket water around cylinder liners (pump, **radiator or cooling tower**), keeps temp ~ 70-85 deg C
Lubrication system	Oil sump, pump, filter, cooler; reduces friction and wear and cools the piston; **wet sump / dry sump**
Starting system	Electric starter motor and battery (small); **compressed air** at 20-30 bar (large)
Governing system	Mechanical / electronic governor controls fuel to hold the speed against load changes

Fuel storage

- Diesel (HSD, LDO, heavy fuel oil) is stored in **underground or above ground steel tanks**, with enough for **7-15 days** run; a smaller **day tank** (about 8 hours) is kept near the engine and fed by gravity.
- Precautions: fire safety, tank vent, **dike (bund wall)** for leakage, settling of water and sludge (drain cock), heating of heavy oils for pumping, fuel quality (**cetane number 45-55**, viscosity, sulphur, flash point).
- Fuel **filters** (primary + secondary) are important as injector clearances are tiny.

Fuel injection

- **Air injection** (obsolete) and **solid injection**: **common-rail**, unit injector, pump-line-nozzle systems. A good spray needs: atomisation, penetration, uniform distribution.
* Plant efficiency (overall) 30–38%. Heat balance of a diesel engine (approx.): useful work 35%, exhaust gas 30%, jacket cooling water 30%, friction and radiation 5%.

Waste heat

- The exhaust (350–450 deg C) is passed through a **waste-heat boiler** to raise steam or hot water -> improves overall utilisation (**cogeneration**, 60–70%).

Supercharging and governing

- **Supercharging** raises inlet air pressure (1.2–2.5 bar) so more air (oxygen) enters -> more fuel burnt -> **power up 30–50%**. Types: **turbocharger** (exhaust-gas driven turbine, most common), mechanical (gear-driven blower), with an **intercooler / after-cooler** for denser air.
- **Governing:** quantity governing (change fuel quantity per cycle, **used in diesel**) -- as the load rises speed falls, the governor increases fuel to restore rated speed. Parallel operation needs governor **droop 3–5%** for load sharing.

Cooling and lubrication in detail

- **Cooling:** about 30% of heat goes to the jacket water. Open (thermosyphon), closed with **radiator + fan**, or cooling-tower system; coolant treated with corrosion inhibitors; thermostat keeps 70–85 deg C. Too cold -> corrosion, poor combustion; too hot -> seizure, oil breakdown.
- **Lubrication:** reduces friction (**30% of mechanical loss**), cools, seals the piston rings, cleans, and protects against corrosion. Oil pressure 3–5 bar; filter, cooler and centrifuge keep it clean.
- **Starting:** electric motor (up to ~ 100 kW), compressed air (valve opens air into cylinders in firing order), or **air motor**. Black-start is possible, important in grid failure.

Selection of site and size

- Near the **load centre**, fuel supply route, small water source, away from residential areas (noise) with room for tank farm; foundations on hard soil.
- Number of units: several **equal-size sets** (n + 1 with one spare) give better part-load efficiency and reliability.

Typical Indian applications

- Remote / island electrification, **telecom towers**, hospitals and data centres (UPS + DG), construction sites, **captive power** in industries, **peak** shaving, and as **black-start units** for big power stations.

4. Performance and Examples of Diesel Plants

Example 1: Diesel set

- A diesel engine uses 800 kg of fuel per hour (CV = 42,000 kJ/kg) and delivers 3200 kW of brake power.
- Brake thermal efficiency = 3200 x 3600 / (800 x 42,000) = 11.52×10^6 / 33.6×10^6 = **0.343 = 34.3%**
- sfc = 800 / 3200 = **0.25 kg / kWh**

Example 2: Cost per unit

- 2 MW set, load factor 50%: energy per year = 2000 x 0.5 x 8760 = **8.76×10^6 kWh**; diesel at Rs 90 / kg with sfc 0.25 -> fuel cost = **Rs 22.5 / kWh**; add maintenance and fixed charges -> Rs 25-30 per kWh -- expensive, hence used for stand-by and peaking only.

5. Gas Turbine Plant - Introduction and Classification

- **Gas turbine (GT) plant** uses hot, high-pressure **gas** (products of combustion) to rotate a turbine and drive an alternator. It works on the **Brayton (Joule) cycle**.
- Basic parts: **compressor, combustion chamber, turbine** (+ starting motor, controls, auxiliaries).

Classification

- **By cycle:** (1) **Open cycle** -- air taken from atmosphere, exhaust released. (2) **Closed cycle** -- the same working fluid (air, helium) circulates; heat is added externally. (3) Semi-closed.
- **By combustion:** constant-pressure (**most**) and constant-volume (explosion - obsolete) type.
- **By use:** industrial / power (heavy-duty frame, e.g. 100-400 MW class), **aero-derivative** (light, quick), micro-turbines (< 500 kW), marine, and locomotive.
- **By compressor:** axial flow (large) or centrifugal (small).

6. Construction of Gas Turbine Plant

- **Compressor:** axial-flow multi-stage (pressure ratio 10-30), driven by the turbine itself (**compressor takes ~ 50-60% of the turbine work**). **Centrifugal** for small ones.
- **Combustion chamber:** can, annular or can-annular; air divided into **primary** (burns fuel), **secondary** (completes combustion) and **tertiary / dilution air** (cools gas to turbine inlet temperature of **1100-1500 deg C**). Fuel: natural gas, diesel, kerosene.
- **Turbine:** axial flow, 2-4 stages, blades from nickel super-alloys with **film cooling** and thermal barrier coating. Speed 3000 / 3600 rpm.
- **Auxiliaries:** starting motor / diesel engine, **lubricating oil system**, fuel system, air filter, **inlet air cooler / fogging**, silencer, **governing** (fuel control), generator, exhaust diffuser + stack, and the **regenerator / heat exchanger** (if fitted).

7. Layout of Gas Turbine Plant with Auxiliaries and Working Principle

Open cycle gas turbine

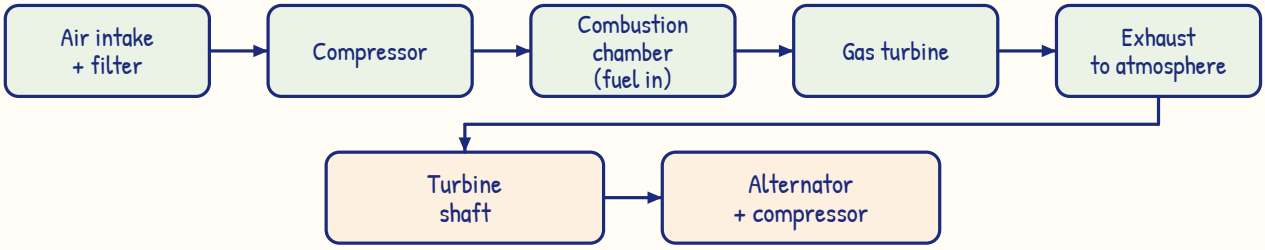


Fig: Open-cycle gas turbine plant

- Air is compressed (1-2), heated by burning fuel at constant pressure (2-3), expanded in the turbine (3-4) and exhausted. **Net work** = turbine work - compressor work.

Closed cycle gas turbine

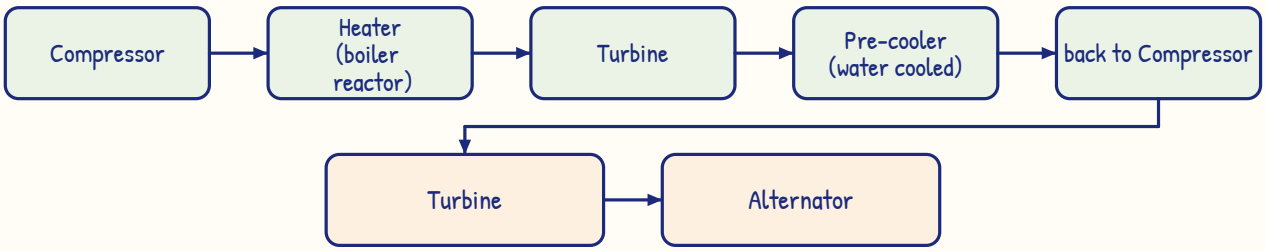


Fig: Closed-cycle gas turbine plant

- Working fluid (air, helium, CO_2) circulates in a closed loop; heat is supplied in a **heater** from external fuel, even **coal or nuclear** (high-temperature gas-cooled reactor); **pre-cooler** rejects heat. Can use **high pressure** (smaller size, better heat transfer).

8. Brayton Cycle Analysis

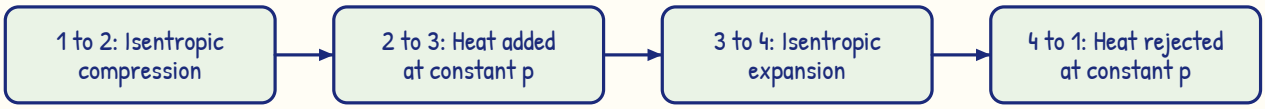


Fig: Ideal Brayton cycle processes

Formulae (ideal air standard)

- Pressure ratio $r_p = p_2 / p_1$; $T_2 / T_1 = r_p^{(\text{gamma}-1)/\text{gamma}}$; T_3 / T_4 = same
- Compressor work $W_C = c_p (T_2 - T_1)$; Turbine work $W_T = c_p (T_3 - T_4)$; Net work $W = W_T - W_C$

- Heat added $Q_1 = c_p (T_3 - T_2)$. **Thermal efficiency = 1 - 1 / $r_p^{(gamma-1)/gamma}$**
- **Work ratio** = W_{net} / W_T (0.3–0.6 for GT; bad if compressor / turbine efficiencies are low)
- Optimum pressure ratio for **max. work**: $r_p = (T_3 / T_1)^{gamma / (2(gamma-1))}$

- Efficiency depends only on pressure ratio. Specific work depends on **turbine inlet temperature (TIT)**. Real cycle: isentropic efficiencies of compressor eta_c (0.85) and turbine eta_t (0.88).

Methods to improve performance

- **Regeneration (heat exchanger):** exhaust heat preheats compressed air -> saves fuel; effectiveness 0.6–0.8; useful at low pressure ratio.
- **Intercooling:** cool air between compressor stages -> less compressor work.
- **Reheating:** gas reheated between turbine stages -> more turbine work.
- Use of all three approaches Ericsson cycle; but adds complexity and pressure drop.

Example 3: Brayton cycle

- Air at 1 bar, 300 K compressed with $r_p = 8$, turbine inlet 1200 K. gamma = 1.4, c_p = 1.005 kJ/kg K.
- $T_2 = 300 \times 8^{0.2857}$ = **543.4 K**; T_4 = 1200 / 1.811 = **662.6 K**
- W_C = 1.005 (543.4 - 300) = **244.6 kJ/kg**; W_T = 1.005 (1200 - 662.6) = **540.1 kJ/kg**; W_{net} = **295.5 kJ/kg**
- Efficiency = 1 - 1/1.811 = **0.448 (44.8%)**; Work ratio = 295.5 / 540.1 = **0.547**

Example 3b: Regenerative Brayton cycle

- Same data as Example 3 (T_2 = 543.4 K, T_4 = 662.6 K) with a regenerator of **effectiveness 0.7**.
- Air leaving regenerator $T_x = T_2 + 0.7 (T_4 - T_2)$ = 543.4 + 0.7 x 119.2 = **626.8 K**

- Heat added = 1.005 (1200 - 626.8) = **576.1 kJ/kg** (earlier 659.9); Efficiency = 295.5 / 576.1 = **0.513 (51.3%)** -- better than 44.8%.

Aero-derivative and heavy-duty turbines

- **Heavy-duty (frame) GT:** thick casing, long life, 50-500 MW, lower efficiency in simple cycle (33-40%), used in CCGT. Examples of frame classes: E, F, H, J.
- **Aero-derivative GT:** based on jet engines; light, compact, **start in 2-5 min**, 20-100 MW, efficiency 35-42%; for peaking, offshore, ships.
- Typical TIT: older 900 deg C -> F-class 1400 deg C -> H/J class **1600 deg C** (advanced cooling + coatings); **ISO rating** = 15 deg C, 1.013 bar, 60% RH.

Effect of ambient conditions

- Output drops about **0.7-1% for every 1 deg C rise** in ambient temperature -> use **inlet air chillers / evaporative cooling / fogging**.
- Efficiency also reduced by altitude and by **compressor fouling** (water wash), inlet and exhaust pressure drops.

Fuels for GT

- Natural gas (preferred, clean), naphtha, kerosene, HSD, syngas (IGCC), biogas, and **hydrogen blends**. Liquid fuel needs good atomisation and clean (no vanadium / sodium) to avoid hot corrosion of blades.

Gas turbine governing and control

- Fuel flow is varied to hold speed (frequency); controls: **temperature limit** (TIT), acceleration limit, compressor **surge protection** (bleed valves, variable inlet guide vanes).
- **Starting sequence:** purge -> crank by starting motor (or static frequency converter) to ~ 20% speed -> ignite -> accelerate -> self-sustaining -> synchronise.

9. Advantages and Disadvantages of Gas Turbine Plants

Advantages	Disadvantages
Small size, light weight, simple layout	Low efficiency (open cycle simple: 25-35%)
Quick start (5-10 min); good for peak load	Compressor takes large part of turbine work
Little cooling water required	High-grade fuel (gas, distillate) is costly
Low capital cost per kW and short installation time	Blades face very high temperature -> special alloys
Smooth running, less vibration, low lubrication oil use	Short life, noise (intake + exhaust), part-load efficiency poor
Wide fuel flexibility, easy automatic operation	Output falls on hot days (air less dense) and at altitude

Point	Diesel plant	Gas turbine plant	Steam plant
Size (MW)	up to 50	up to 400	up to 1000+
Start-up	1-2 min	5-10 min	hours
Efficiency	35-40%	30-40%	35-45%
Water needed	low	very low	very high
Fuel	HSD / HFO	gas / distillate	coal etc.
Use	stand-by, peak	peak, combined cycle	base load

10. Combined Cycle Power Plants

- ***Combined cycle (CCGT)*** = a ***gas turbine (Brayton, topping cycle)*** + a ***steam turbine (Rankine, bottoming cycle)***. The hot exhaust (500-600 deg C) of the GT goes to a ***heat recovery steam generator (HRSG)*** to raise steam.

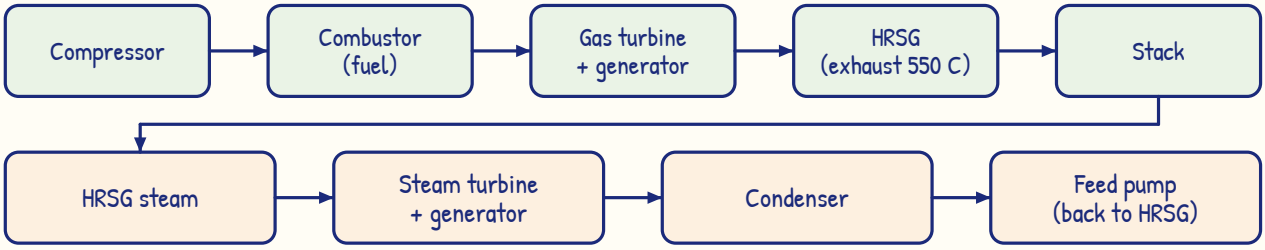


Fig: Combined-cycle power plant

- **Efficiency:** GT 35-40% + ST contribution -> overall **55-62%** (best among thermal plants); about **two-thirds of power from GT** and one-third from ST.
- Plant sizes: 100-1000 MW; unit uses 1 GT + 1 ST (single shaft) or 2 GT + 1 ST (**2 x 1 multi-shaft**).

HRSG

- Parts: **superheater, evaporator, economiser** (counter-flow), drum, **pinch point** temperature difference 10-20 deg C. Types: single, double and triple-**pressure** (HP, IP, LP) with reheater to raise efficiency. Supplementary firing (duct burners) can raise steam output.

Combined-cycle efficiency formula

- $\eta_{cc} = \eta_{GT} + \eta_{ST}(1 - \eta_{GT})$ - losses. Example: $0.36 + 0.33(1 - 0.36) = 0.36 + 0.211 =$ **0.571 = 57%**

Advantages

- **Highest thermal efficiency**, low fuel use, **low CO_2 per kWh** (about 40-50% less than coal), very low SO_x, lower water need than steam plant, quick installation (**2 years**), flexible, modular, low capital cost / kW, can be started in 30-60 minutes.

Disadvantages

- Needs clean **gas fuel** (gas supply is uncertain in India), efficiency drops at part load and on high ambient temperature, complex controls, NO_x control needs **DLN burners** or water / steam injection.
- Other forms: **IGCC** (coal gasification + combined cycle), **Integrated solar combined cycle**, cogeneration with process steam.

Applications of GT and CCGT

- Peak power, base load (CCGT) where gas is available, **gas pipelines**, offshore platforms, aircraft / ship propulsion, **cogeneration** in refineries / paper / fertiliser, power supply on trains and emergency standby.

II. Important Short Questions (exam revision)

Q & A

- **Why is a diesel plant used as stand-by?** Quick start, no stand-by loss, small size; but running cost is high.
- **What is the compression ratio of a diesel engine?** 14:1 to 22:1 (higher than petrol engine, as only air is compressed).
- **Why is a silencer used?** Exhaust gas leaves at high pressure and velocity; silencer reduces noise and pulsations.
- **Why is a governor needed?** To keep speed (and so frequency, 50 Hz) constant by varying the fuel supplied as the load changes.
- **What is surge in a GT compressor?** Reverse flow at low mass flow / high pressure ratio causing violent vibration; prevented by bleed valves.
- **Why does a GT compressor consume 50-60% of turbine work?** Gas is compressed in a cold state; gases expand hot with a large volume, so turbine produces much more work.
- **Topping and bottoming cycle?** Topping (Brayton, high temperature) rejects heat at ~ 550 deg C to the bottoming (Rankine, low temperature).
- **What is the pinch point in an HRSG?** Minimum temperature difference between exhaust gas and water / steam at the evaporator inlet; small value gives more steam but bigger area.
- **Closed cycle advantage?** Clean working fluid (no blade fouling), any fuel, high pressure -> compact; but needs a heat exchanger (large) and cooling water.

Quick revision

- Diesel plant: quick start, 35–40% efficiency, stand-by / peak use. 4-stroke vs 2-stroke. Systems: fuel, air, exhaust, cooling, lubrication, starting.
- Brayton efficiency = $1 - 1 / r_p^{(\text{gamma}-1)/\text{gamma}}$. Improve by regeneration, intercooling, reheating.
- Open cycle: air + fuel, exhaust to atmosphere. Closed cycle: working fluid recirculated, external heater.
- Combined cycle = GT + HRSG + ST, efficiency 55–60%.